

You-Ting Wu (*he/him*)

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Research Interests

climate modeling, Arctic climate attribution and risk assessment, machine learning for climate, energetic feedbacks

Education

Columbia University

Ph.D., Applied Mathematics and Climate Science

📅 Sept 2024 – present 📍 New York, NY, USA

National Taiwan University

Double B.Sc., Mechanical Engineering & Atmospheric Sciences

📅 Sept 2018 – July 2023 📍 Taipei, Taiwan

Skills & Awards

Skills: Time Series, Nonlinear Dynamical Systems, High-Performance Computing, Numerical Analysis, Python (PyTorch, SciPy, Scikit-learn), C/C++, MATLAB

Awards:

- Government Scholarship to Study Abroad, Ministry of Education, Taiwan (Jun 2026)
- Nominated for Membership of $\Sigma\Xi$ (The Scientific Research Honor Society) (Jan 2026)
- Excellent Undergraduate Research Project Award, Ministry of Science and Technology, Taiwan (Jun 2023)
- Undergraduate Research Project Grant, Ministry of Science and Technology, Taiwan (Jun 2022)

Professional Experience

Graduate Research Assistant, Columbia University

📅 Sept 2024 – present 📍 New York, NY, USA

- Modeled nonlinear, bifurcating dynamics of the Arctic climate system under CO₂ forcing using large-ensemble Earth-system-model simulations; identified a critical tipping threshold via regime-transition analysis
- Built CNN and LSTM neural-network emulators to predict spatiotemporal climate-risk indicators (cold-spell extremes) from Earth-system-model output; evaluated out-of-sample generalization across unseen forcing scenarios
- Quantified climate-risk metrics (sea-level rise, Arctic sea-ice loss) under differing anthropogenic forcings (CO₂ vs. aerosols) using coupled climate and numerical models

Engineering Intern, Formosa Climate Smart Service Limited

📅 July 2023 – Sept 2023 📍 Taipei, Taiwan

- Built an ETL data pipeline converting raw engineering inputs into quantitative Energy Use Intensity (EUI) metrics across large building datasets
- Developed a Python-based analytics/visualization platform; automated geospatial and CAD data integration into a 3D building-EUI workflow

Visiting Student, Lamont-Doherty Earth Observatory

📅 June 2022 – Aug 2022 📍 Palisades, NY, USA

- Applied bootstrapping resampling and statistical significance testing to large-ensemble climate simulations to quantify the robustness of an aerosol-driven Arctic amplification signal against internal variability

Publications

5. Previdi, M., **Wu, Y.-T.**, Polvani, L.M., England, M.R., Sigmond, M., Liang, Y.-C. Substantial Twentieth-Century Global Sea Level Rise Caused by Ozone-Depleting Substances. Submitted to *Journal of Climate* (under revision).
4. **Wu, Y.-T.**, Previdi, M., Romanou, A., Lerner, P., Polvani, L.M. Arctic Climate Tipping Points under Delayed CO₂ Emissions Mitigation. Accepted by *Communications Earth & Environment* (to appear).
3. Liang, Y.-C., Polvani, L.M., Previdi, M., Dong, Y., **Wu, Y.-T.**, England, M.R., Griffith, S.M. (2026). The Importance of Non-CO₂ Greenhouse Gases to Arctic Warming and Sea-Ice Loss. *Geophysical Research Letters*, 53, e2026GL122867.
2. **Wu, Y.-T.**, Liang, Y.-C., Previdi, M., Polvani, L.M., England, M.R., Sigmond, M., Lo, M.-H. (2024). Stronger Arctic Amplification from Anthropogenic Aerosols than from Greenhouse Gases. *npj Climate and Atmospheric Science*, 7(1), 142.
1. **Wu, Y.-T.**, Liang, Y.-C., Kuo, Y.-N., Lehner, F., Polvani, L.M., Previdi, M., Lo, M.-H., Lan, C.-W. (2023). Exploiting SMILEs and the CMIP5 Archive to Understand Arctic Climate Change Seasonality and Uncertainty. *Geophysical Research Letters*, 50(2), e2022GL100745.

Presentations & Teaching

12x scientific presentations at international conferences (2021–2025)

10x teaching assistantships, including Partial Differential Equations, Numerical Analysis, and Applied Mathematics, at Columbia University and National Taiwan University